Performance Evaluation of Computer Systems and Networks
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Fragmenting Wireless Link

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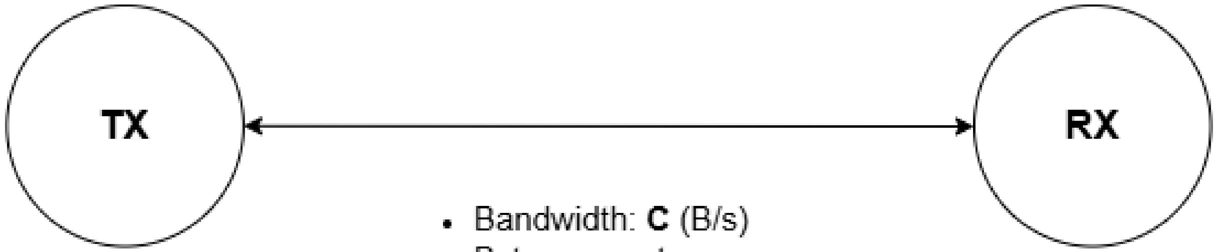
Introduction

- **Transmitter** divides packets into fragments of size **P** and adds a header of size **H**
- **Wireless link:** fragments of size **F** = H + P are transmitter at **C** B/s with ***a*** byte error rate
- **Receiver** considers packets corrupt with probability $\boldsymbol{p} = 1 – (1 - \alpha)^F$ and responds with ACK/NACK



- Bandwidth: **C** (B/s)
- Byte error rate: **α**

- Packet size: **S**min - **S**max (B)
- Packet Interarrival time: E[**T**] (s)
- Fragment size: **P** (B)
- Header size: **H** (B)

- Ack size: **A** (B)

- Fragment with header: **F** (B)
- Corrupt packet probability: ***p*** = 1 - (1 - α)^F

Objectives

- Find optimal value of P that minimizes:
  - End-to-End delay of reassembled packets
  - System utilization
- Estimate optimal buffer size of real transmitter
- Assess fairness of system

Key performance indices

- End-to-end packet delay
- Fragments in queue
- System utilization

Factors

- **P** – Fragment size
- **α** – Byte Error rate

Model, Assumption and Implementation

Model

- The system is modeled as a M/G/1 system with group arrivals due to fragmentation

Assumptions

- Fragmentation and reassembly delays are negligeable
- System has infinite queue

Implementation

- OMNET++
- Simple modules: Tx and Rx
- Custom messages exchanged using sendDirect()

Scenarios

Bluetooth Industrial

- Context: Predictive maintenance (Vibration analysis)
- Traffic: Large packets (FFT data), High Noise environment

Bluetooth Audio Streaming

- Context: Streaming (AAC)
- Traffic: Medium packet size, Low Noise

Bluetooth Sensors

- Context: Wearable IMUs
- Traffic: Small packets, Medium-High Noise (Body shadowing)

Verification and Validation

Internal Consistency:

- Little's Law: $E[N] = \lambda E[R]$ and $E[Nq] = \lambda E[W]$.
- Utilization: $\rho = \lambda E[ts]$
- Additivity of mean: $E[R] = E[W] + E[ts]$ and $E[N] = E[Nq] + \rho$.
- Flow consistency: $\gamma = \lambda$ (throughput equals arrival rate)

Degeneracy Tests:

- $\alpha = 0$: no retransmissions
- $\alpha = 1$: zero throughput
- P = 10kB: zero throughput

Consistency:

- Double packet size and double inter-arrival time: same throughput

Continuity:

- Small input perturbations: inter-arrival time, packet size, byte error rate
- Small output variations: mean delay, throughput

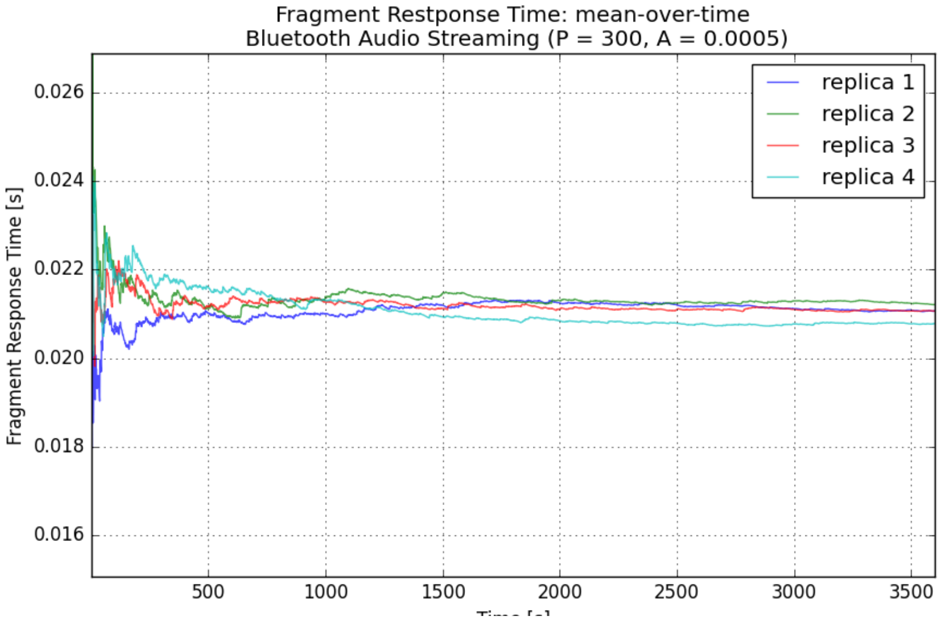
Discrepancies with theoretical model:

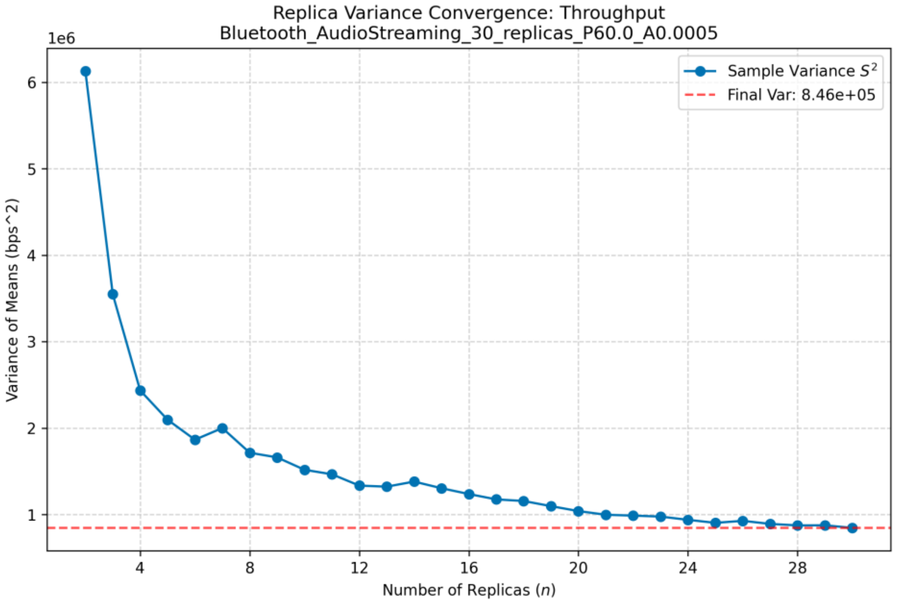
- Errors < 5% in scenarios with high fragmentation
- Exception: Sensors scenario (mean fragments per packet = 1.5)
  - Error up to 70% in waiting time
  - Why? Head-of-Line blocking: full-sized fragments block small tail fragments
  - Theoretical model assumes "average" fragments, missing heterogeneity

Data Analysis Methodology

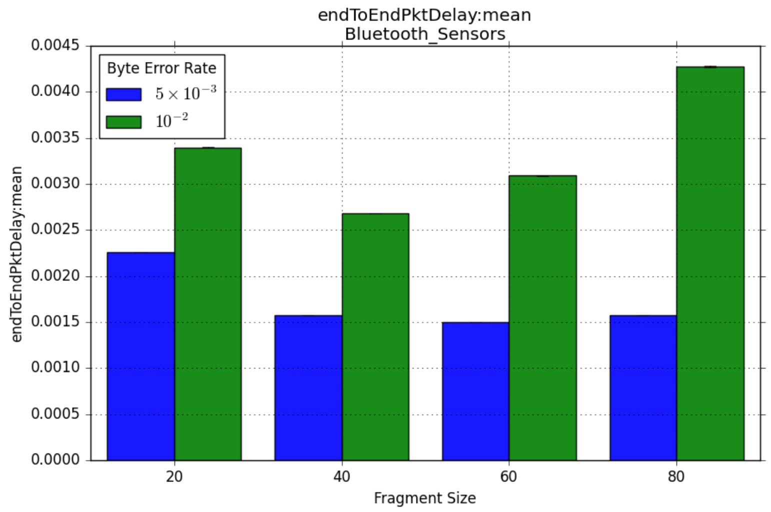
Statistical robustness

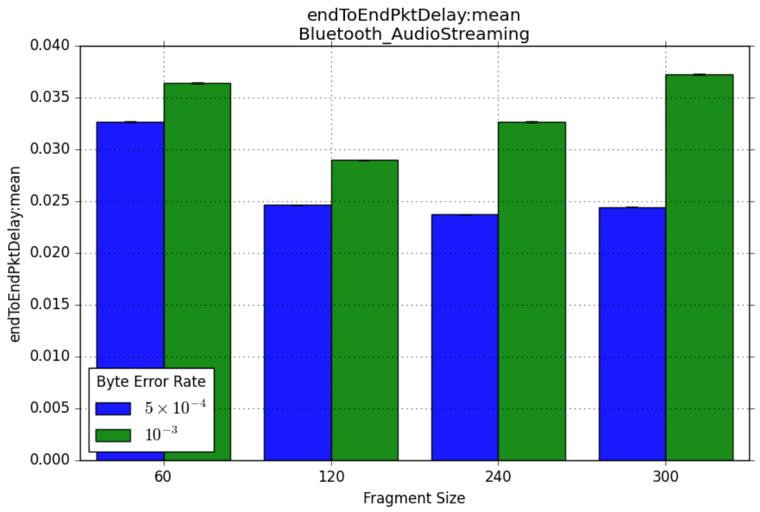
- 30 replicas with different RNG seeds
- 3 separate RNG for each simulation
- 95 % confidence intervals
- Warm-up time estimation
- Variance convergence check

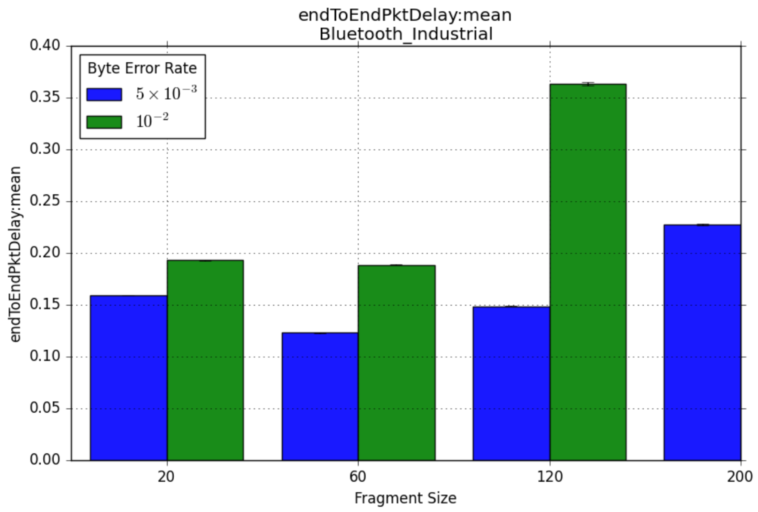




End-to-End Delay (1/2)







Results:

- Expected ‘U’ shape

Utilization:

- Analog results

Before saturation:

- Header and protocol overhead dominate

At saturation:

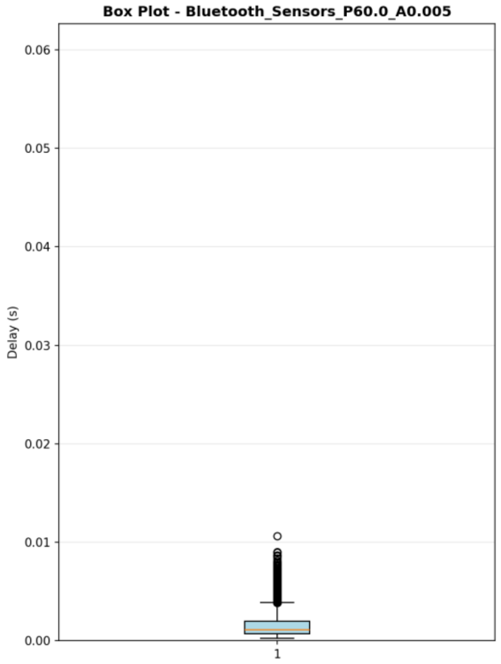
- Optimal balance

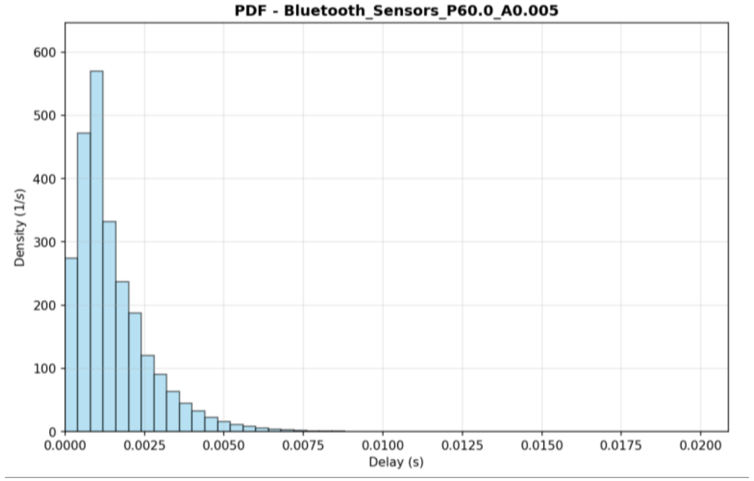
After saturation:

- Retransmissions dominate

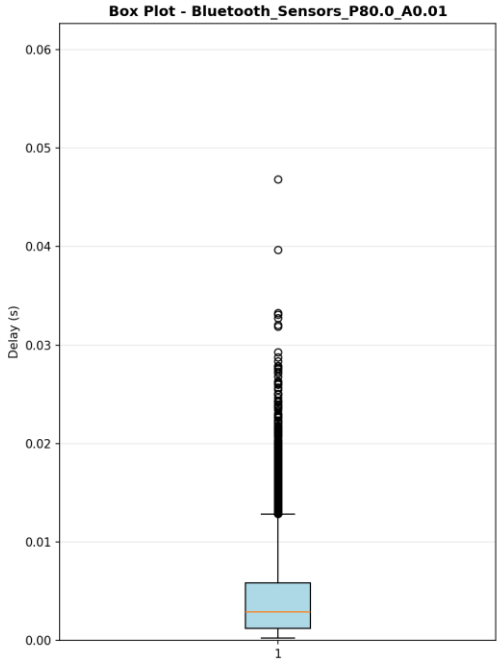
End-to-End Delay (2/2)

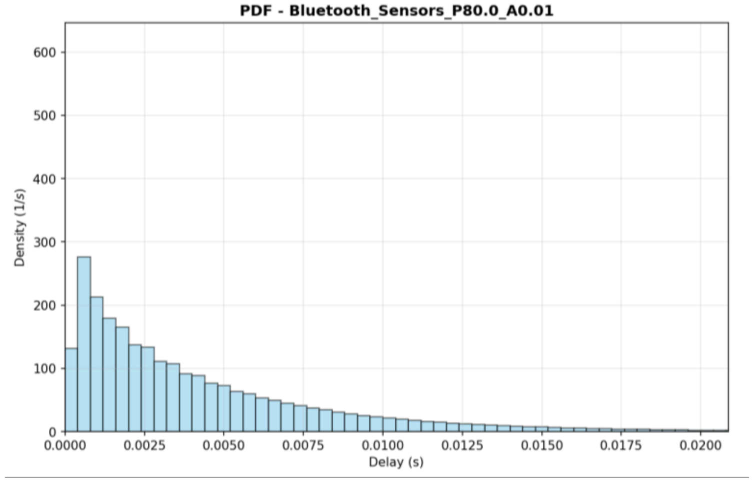
Sensors: optimal low noise





Sensors: worst high noise





Fairness and Predicatibility

Insight:

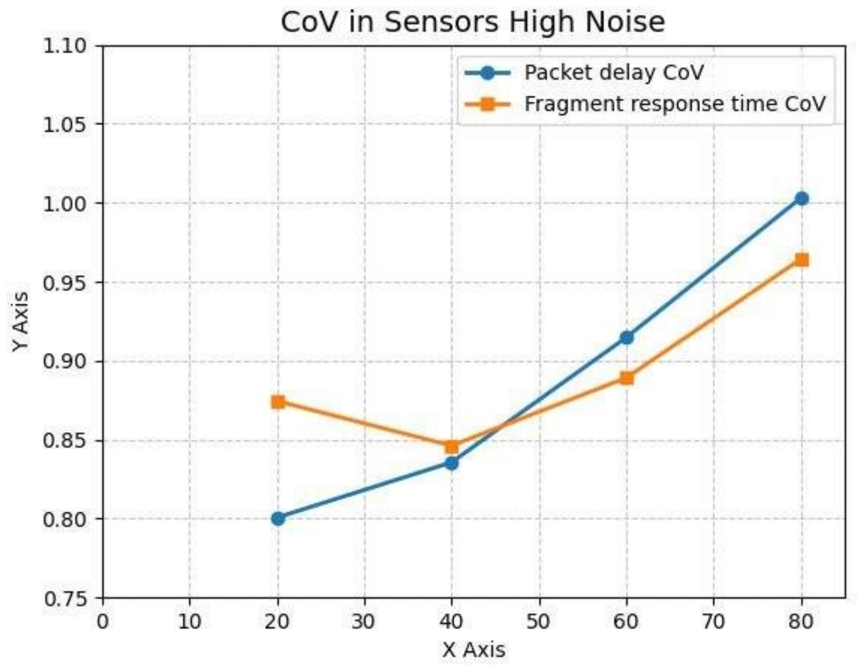
- Trade-off: optimizing protocol efficiency through increased fragment size (P) beyond saturation inevitably compromises system delay predictability
- More accentuated in noisy environments.

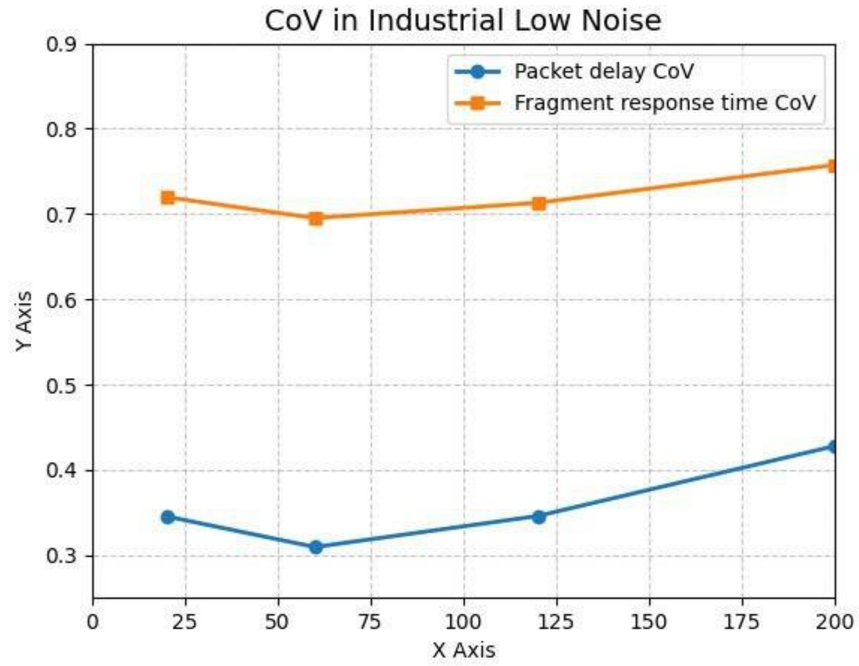
Sensors:

- Packet level predictability (Cov) and fairness (Gini) always decrease with increasing P
- Small Fragment number: no statistical smoothing

Industrial and Streaming:

- U shape predictability and fairness with increasing P
- Large Fragment number: statistical smoothing





Buffer Size Estimation

Bluetooth Industrial

- 50 – 80 kB
- Feasible for modern sensors
- Legacy or Ultra-low-power may experience buffer overflow

Bluetooth Audio Streaming

- 30 – 40 kB
- Stable queue size easily manageable by audio streaming applications

Bluetooth Sensors

- 2 – 4 kB
- Reference sensors would experience no buffer space issues

Table 2: Global Peak Queue Depth Analysis across scenarios

Scenario	P	α	$N_{q,\max}$	**Queue dimension (KB)**
Sensors	20	0.005	68	1.3
	20	0.010	108	2.1
	40	0.005	35	1.4
	40	0.010	52	2.0
	60	0.005	24	1.4
	60	0.010	41	2.4
	80	0.005	26	2.0
	80	0.010	50	3.9
Audio Streaming	60	0.0005	480	28.1
	60	0.0010	622	36.4
	120	0.0005	212	24.8
	120	0.0010	253	29.6
	240	0.0005	115	27.0
	240	0.0010	145	34.0
	300	0.0005	103	30.2
	300	0.0010	116	34.0
Industrial	20	0.005	2245	43.8
	20	0.010	2169	42.4
	60	0.005	640	37.5
	60	0.010	907	53.1
	120	0.005	369	43.2
	120	0.010	614	72.0
	200	0.005	249	48.6
	200	0.010	–	–

Conclusions

U-shaped performance:

- Optimal P balances protocol overhead vs. retransmissions

Noise matters:

- High-error environments shift optimum to smaller P

Fragmentation regime matters:

- Low fragmentation: Protocol efficiency gains come at the cost of delay predictability